DE-CLIP: Few-Shot Anomaly Detection via Difference-Guided Embedding Editing

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Abstract

Anomaly detection (AD) plays a critical role in applications such as automated industrial inspection and medical image analysis. Empowered by the strong pre-trained vision-language model, CLIP, recent years have witnessed the emergence of several CLIP-based few-shot AD methods. Due to the overlap between the embedding distributions of normal and anomalous samples, many existing approaches introduce additional model training for more discriminative text embeddings. However, we demonstrate that such training is not necessary. Specifically, we find that this embedding overlap can be separated by introducing a Difference-guided vector for embedding Editing (DiffEdit). Based on this finding, we propose DE-CLIP, a simple yet effective framework based on DiffEdit, which directly edits text embeddings based on the textual and visual differences between normal and anomalous samples, resulting in more discriminative embeddings for AD. Extensive experiments on industrial and medical datasets demonstrate the superiority of our proposed DE-CLIP compared with existing baselines. For instance, on MVTec dataset, DE-CLIP achieves 96.6% and 96.7% AUROC on anomaly classification and segmentation, surpassing both training-based and training-free methods. In addition, we observe that introducing DiffEdit into other training-free baselines could also significantly improve their performance, highlighting the potential of DiffEdit to promote better AD.

1 Introduction

Anomaly detection (AD) is a critical task in safety-sensitive domains such as industrial inspection and medical diagnostics, where rare but significant deviations from normality should be identified accurately. AD typically comprises two sub-tasks: anomaly classification (AC), which distinguishes abnormal from normal instances, and anomaly segmentation (AS), which localizes the abnormal regions at the pixel level (Bergmann et al., 2019; Fernando et al., 2021). However, these tasks remain highly challenging in practice due to the scarcity of anomalous samples, leading to long-tailed data distributions and insufficient training signals (Liang et al., 2022; Tyshchuk et al., 2023; Schrodi et al., 2024). This imbalance creates a barrier to both learning and generalization, especially under zero- or few-shot settings (Song et al., 2022; Li et al., 2024b; Zhang et al., 2024).

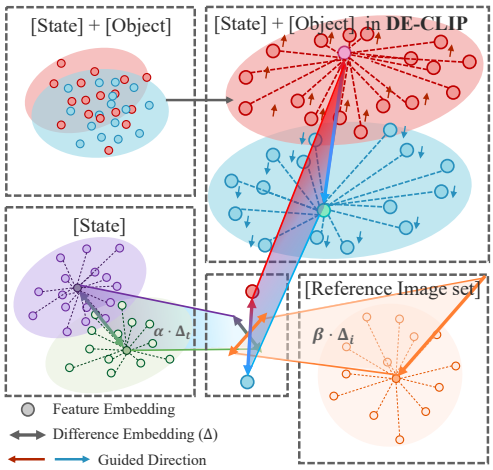

Figure 1: Overview of DE-CLIP 's DiffEdit guided embedding mechanism. Feature embeddings are steered using difference vectors Δ derived from textual and visual domains

Recent advances in vision-language models (VLMs), especially the contrastive language–image pretraining (CLIP) model (Radford et al., 2021), offer a promising solution to these challenges. CLIP (Radford et al., 2021), trained on approximately 400 million noisy image-text pairs crawled from the internet, exhibits remarkable zero-shot generalization capabilities, particularly excelling in image classification and recognition tasks (Qu et al., 2024; Li et al., 2021, 2022, 2023; Radford

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et al., 2021). However, its performance on anomaly understanding remains limited. A key bottleneck lies in the semantic embedding space of CLIP's text encoder: prompts with semantically opposite meanings (e.g., *intact bottle* vs. *broken bottle*) are often mapped into nearby regions (Ma et al., 2025; Eslami and de Melo, 2024; Kang et al., 2025; Tong et al., 2024; Tang et al., 2023; Suo et al., 2023), resulting in decision ambiguity and blurred boundaries between normal and anomalous concepts.

WinCLIP (Jeong et al., 2023) is an early CLIP-based AD method that constructs extensive handcrafted prompts and performs multiple forward passes for zero-/few-shot classification and segmentation. VAND (Chen et al., 2023) extends this with projection learning from reference images. More deeply integrated approaches—such as AdaCLIP (Cao et al., 2024), AnomalyCLIP (Zhou et al., 2023), and AA-CLIP (Ma et al., 2025)—rely on fine-tuning, specialized prompts, or added modules to improve performance. However, these approaches typically rely on large amounts of task-specific training data before being transferred to new datasets. This not only constrains their generalizability and applicability in zero-shot scenarios but also risks disrupting the internal representational space of CLIP.

In this work, we observe that structured antonymic adjective pairs naturally encode interpretable semantic directions within CLIP's embedding space. These directions define a cognitive opposition axis, which we find can be leveraged to enhance conceptual separability. Inspired by psycholinguistic theories of semantic differentials (Lakoff and Johnson, 1980; Langacker, 1987; Barsalou, 1999), we propose to inject directional difference vectors into CLIP's latent space. Surprisingly, by aligning these textual shifts with representative visual anchors, the resulting edited embeddings become more grounded and disentangled, leading to clearer semantic boundaries for AD tasks.

Building on this insight, we propose DE-CLIP: a plug-and-play, training-free framework for few-shot AC and AS. DE-CLIP operates by computing a semantic difference vector from antonymic state word pairs, and injecting it into state-object prompts. Notably, while DE-CLIP requires no parameter updates or fine-tuning, it does leverage a small number of normal samples and a predefined antonym lexicon to guide inference. Thus, "training-free" in our context refers strictly to the absence of gradient-based optimization. This process creates a perceptual-semantic axis that systematically transforms the embedding space. To further ground these edits, we align the semantic axis with visual reference embeddings from normal samples. As shown in Figure 1, this dual-modality difference guided editing introduces structured tension between normal and anomalous prompts. We call this adjustment method Difference-guided embedding Editing (DiffEdit). On industrial and medical datasets, DE-CLIP achieves the AUROC of 96.6%/96.7%, 87.1%/97.6%, and 75.2%/92.6% for AC/AS on MVTec, VisA, and Brain MRI datasets, respectively, outperforming existing baselines, including both training-based and training-free methods, and has great potential to enhance the performance of other training-free ones.

Overall, our contributions are four-fold.

- We identify semantic overlap in CLIP's text embedding space as a key bottleneck for anomaly detection and formulate a solution by introducing a Difference-guided vector for embedding Editing (DiffEdit).
- We propose DE-CLIP, a dual-guided editing framework that introduces semantic difference vectors from both textual and visual modalities to separate semantic overlap in the embedding space without training.
- We conduct extensive evaluations on multiple industrial and medical AD datasets and show that DE-CLIP consistently outperforms existing approaches in both AC/AS.
- DiffEdit can also improve its performance when combined with other baselines, demonstrating the potential of Difference-Guided Embedding Editing to promote better AD. DiffEdit yields up to a 30.1% increase in AUROC.

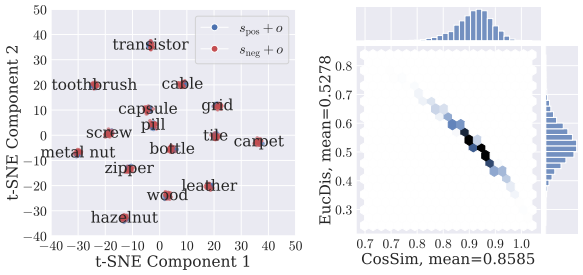
2 Problem Formulation

2.1 Zero/Few-Shot Anomaly Detection

Problem Set-Up. Zero-shot AD separates normal from anomalous samples without any defect labels, while few-shot AD augments each object class with K reference normals to improve robustness in scarce-data regimes.

Given a test image $x \in \mathcal{X}$ and prompts $\{t_i\}_{i=1}^{N}$, CLIP embeddings are

$$E(I) := f_{\text{img}}(x), \qquad E(T) := f_{\text{text}}(t_i).$$



(a) (b)

Figure 2: (a) t-SNE visualization of composed embeddings $E(S+O)$. (b) Cosine similarity and Euclidean distance between normal and anomalous $E(S+O)$ embeddings.

Zero-shot uses prompts alone; few-shot adapts $E(T)$ with features from the reference set $\mathcal{R}_o = \{x_1^{(o)}, \ldots, x_K^{(o)}\}$.

Binary Test in Joint Space. Define the hypotheses

$$\mathcal{H}_0 : x \sim \mathcal{D}_{\text{pos}} \quad \text{(normal)},$$
$$\mathcal{H}_1 : x \sim \mathcal{D}_{\text{neg}} \quad \text{(anomalous)},$$

and score an image via maximum cosine similarity:

$$\text{score}(x) = \max_i \cos\big(E(x), E(t_i)\big).$$

Geometry Requirement. Let $E_{\text{pos}} := E(t_{\text{pos}})$, $E_{\text{neg}} := E(t_{\text{neg}})$ be canonical normal/defect embeddings. Reliable AD requires

$$\cos(E_{\text{pos}}, E_{\text{neg}}) \ll 1,$$
$$\|E_{\text{pos}} - E(x_{\text{pos}})\| \ll \|E_{\text{neg}} - E(x_{\text{pos}})\|$$

ensuring a wide inter-class angle and tight normal alignment.

2.2 Imperfect Text Embedding

We first analyze the structure of the embedding space $E(S+O)$ to evaluate how well CLIP separates normal and abnormal prompts. Figure 2(a) shows the t-SNE projection of $E(S+O)$ embeddings 15 object classes on MVTec Dataset, where each dot represents either a positive or negative phrase. We observe that clusters of normal and anomalous state words with object are largely overlapping across all categories.

To quantify this semantic entanglement, we compute all pairwise cosine similarities and Euclidean distances between each normal and anomalous $E(S+O)$ vector. The distribution of cosine similarities and Euclidean distances in figure 2(b)further confirms that the positive/negative phrases are densely mixed in the embedding space, showing no significant margin.

We note that many existing prompt-based CLIP adaptations adopt handcrafted templates such as "a photo of a damaged screw." Although such phrases may not exist verbatim in CLIP's pretraining corpus, we conservatively categorize them as belonging to the same class of $E(S+O)$ representations for experimental consistency.

3 Our Proposed DE-CLIP

3.1 Overview

We propose DE-CLIP, a few-shot AD/AS framework that edits prompts with modality-specific shifts: linguistic Δ_T and visual Δ_I (Figure 3). In classification, weights α, β blend the two directions, and the cosine to the query embedding yields the anomaly score. For segmentation, multi-scale query–reference comparisons generate similarity maps that are fused into the final mask, capturing both global and local defects.

3.2 Difference Based on Language Modality

Guidance Potential of $\Delta_{T(S+O)}$. We test whether text-space directions that join state and object tokens can stand in for explicit visual guidance. For object o with positive and negative state vocabularies $\mathcal{S}_{\text{pos}} = \{s_{\text{pos},m}\}_{m=1}^{M}$ and $\mathcal{S}_{\text{neg}} = \{s_{\text{neg},n}\}_{n=1}^{N}$, the composed semantic offset is

$$\Delta_{T(S+O)} = \frac{1}{M}\sum_m E(s_{\text{neg},m} + o) - \frac{1}{N}\sum_n E(s_{\text{pos},n} + o).$$

All difference vectors are ℓ_2-normalised to remove scale effects noted in Fig. 2.

Figure 5 plots cosine similarities between $\Delta_{T(S+O)}$ and Δ_I. Strong diagonal values show that the composed text offsets reliably align with object-specific visual shifts, while several off-diagonal peaks indicate transferable semantic directions that generalise across categories.

Lightweight Directional Editing via $\Delta_{T(S)}$. Motivated by cross-category generalizability, we explore a simplified, object-agnostic direction $\Delta_{T(S)}$, computed purely from state adjectives:

$$\Delta_{T(S)} = \frac{1}{M}\sum_{m=1}^{M} E(s_{\text{neg},m}) - \frac{1}{N}\sum_{n=1}^{N} E(s_{\text{pos},n}).$$

As shown in Fig. 4, this direction already separates normal and abnormal concepts well, even without object context. Averaged across all categories, its cosine similarity with visual embedding

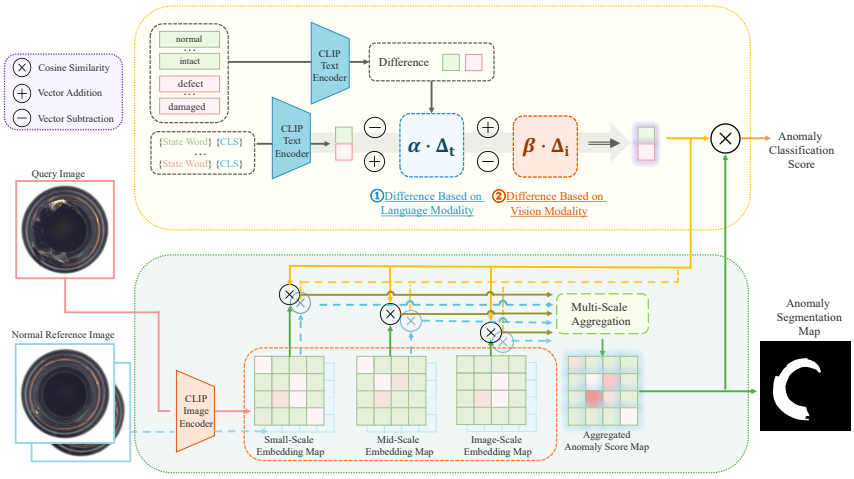
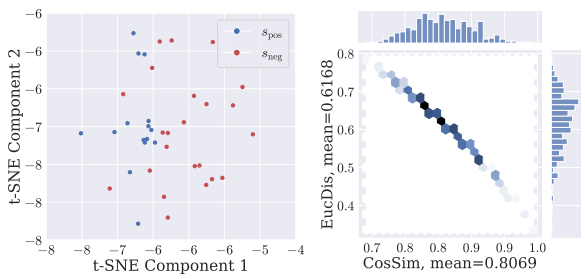


Figure 3: Workflow of our proposed DE-CLIP for few-shot anomaly detection and segmentation.



(a) (b)

Figure 4: (a) t-SNE visualization of state-only embeddings $E(S)$. (b) Cosine similarity and Euclidean distance between normal and anomalous $E(S)$ embeddings.

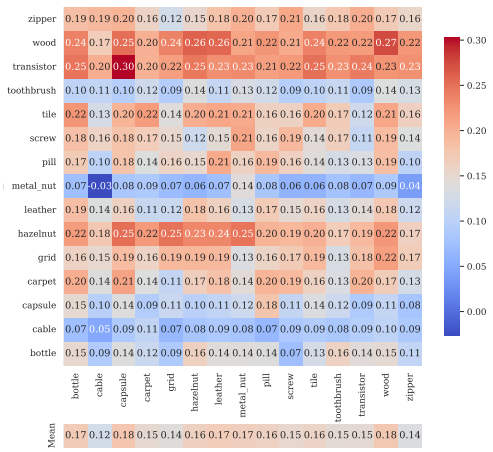


Figure 5: Cosine similarity between Δ_I and $\Delta_{T(S+O)}$ across all object categories. Diagonal values indicate intra-class alignment; off-diagonal values reflect cross-category generalization.

differences reaches 0.14, modest but consistently positive, confirming $\Delta_{T(S)}$'s alignment with visual anomalies. Given its generality and efficiency, we adopt $\Delta_{T(S)}$ as a default editing vector, reducing both computational and annotation overhead.

3.3 Difference Based on Vision Modality

Because language differences capture high-level semantics that subsequently guide low-level visual refinements, we next turn from textual to visual cues to show how the two forms of guidance complement each other.

Similarly, let $\mathcal{I}_{\text{pos}} = \{x_{\text{pos},p}\}_{p=1}^{P}$ and $\mathcal{I}_{\text{neg}} = \{x_{\text{neg},q}\}_{q=1}^{Q}$ represent the collections of positive (normal) and negative (anomalous) image samples.

The average visual embedding difference is

$$\Delta_I = \frac{1}{Q}\sum_{q=1}^{Q} E(x_{\text{neg},q}) - \frac{1}{P}\sum_{p=1}^{P} E(x_{\text{pos},p}). \quad (1)$$

In principle, incorporating visual embedding differences between anomalous and normal samples Δ_I can provide strong semantic alignment between vision and language modalities. Such guidance helps steer the prompt embedding toward regions more consistent with the actual image distribution. However, in practical scenarios, abnormal samples are scarce or unavailable during deployment, limiting access to $\mathcal{I}_{\text{neg}}$.

To address this constraint, we approximate the visual difference using only the normal samples $\mathcal{I}_{\text{pos}}$, we incorporate vision-based guidance through Δ_I, which captures the mean CLIP embedding of a small reference set of normal images,

$$\Delta_I = \frac{1}{P}\sum_{p=1}^{P} E(x_{\text{pos},p}),$$

This vector represents the prototypical visual concept of "normality" and acts as a perceptual grounding signal for guiding semantic contrast.

3.4 Theoretical Insights

Let the class-conditional image means be

$$\mu_+ = \tfrac{1}{P}\sum_{p=1}^{P} E(x_{\text{pos},p}), \qquad \mu_- = \tfrac{1}{Q}\sum_{q=1}^{Q} E(x_{\text{neg},q}),$$

so the visual shift defined in (1) can be written compactly as

$\Delta_T = \Delta_{T(S)}$ and μ_+, μ_- be the class-mean image embeddings. Define

$$\boxed{\Delta_I^{\pm} \;=\; \mu_+ - \mu_-, \qquad \Delta_I^{+} \;=\; \mu_+.}$$

Ideal (Two-Sided) Case. CLIP alignment gives $\Delta_I^{\pm} \approx \delta\Delta_T \quad (\delta > 0)$; under the isotropic assumption, $w^\star \propto \Delta_I^{\pm}$ (Fisher LD optimum(Fisher, 1936)). DiffEdit shifts the positive anchor $\tilde{t}_{\text{pos}} = E(t_{\text{pos}}) + \beta\Delta_I^{\pm}$, hence

$$\tilde{t}_{\text{pos}} - E(t_{\text{neg}}) = \Delta_T + \beta\Delta_I^{\pm} \approx (1+\beta\delta)\Delta_T,$$

i.e. any β rescales but never rotates $w^\star$.

Practical (One-Sided) Case. When only normal images are available, use $\Delta_I^{+} = \mu_+$. Because μ_+ aligns with $E(t_{\text{pos}})$, $\Delta_I^{+} \approx \gamma\Delta_T \;(\gamma > 0)$, so the same derivation yields

$$\hat{t}_{\text{pos}} - E(t_{\text{neg}}) \approx (1+\beta\gamma)\Delta_T,$$

again preserving the FLD direction. Thus DiffEdit maintains optimality even with a small normal reference set.

3.5 DiffEdit: Difference-Guided Vector for Embedding Editing

To mitigate semantic overlap between normal and anomalous prompts in CLIP's latent space, we introduce a directional editing strategy that enhances class separation.

Inspired by Semantic Differential Scales (SDS) (Osgood et al., 1957), which encode meaning along bipolar adjective axes (e.g., good–bad), we construct a task-specific Defect-Oriented Semantic Differential Lexicon comprising antonymic adjective pairs (e.g., intact–broken, stable–cracked). Unlike classical SDS, our lexicon is grounded in domain-relevant visual semantics, ensuring that semantic differences align with perceptually meaningful axes for anomaly discrimination.

Default Difference Vectors. In this work, we use Δ_T and Δ_I^{+} by default to refer to the language- and vision-guided difference vectors, respectively. As discussed previously, Δ_T specifically denotes $\Delta_{T(S)}$. The textual difference vector Δ_T is computed from state word embeddings, while the visual difference vector Δ_I^{+} is derived from a small set of normal reference images.

Dual-Modality Difference-Guided Embedding Construction. Motivated by cognitive semantics (Lakoff and Johnson, 1980; Langacker, 1987; Barsalou, 1999), we apply a structured editing mechanism that integrates both language and vision differences, which we term the Difference-guided vector for embedding Editing (DiffEdit). Specifically,

$$\text{DiffEdit} = \begin{cases} -\alpha\cdot\Delta_T + \beta\cdot\Delta_I^{+} & \text{if } t = t_{\text{pos}} \\ +\alpha\cdot\Delta_T - \beta\cdot\Delta_I^{+} & \text{if } t = t_{\text{neg}} \end{cases}$$

where $s_{\text{pos},m}$ and $s_{\text{neg},n}$ are adjective tokens describing normal and anomalous states, o denotes the object class, and α, β control the strength of semantic and visual guidance. This push-pull formulation ensures that normal prompts are drawn toward visual and linguistic prototypes, while anomalous prompts are repelled along both axes, resulting in an amplified semantic margin.

For clarity, we denote the editing vector as DiffEdit(t), applied conditionally based on polarity.

3.6 AC in DE-CLIP

Prototype Construction. For object class o with normal adjectives $\{s_{\text{pos},m}\}_{m=1}^{M}$ and anomalous adjectives $\{s_{\text{neg},n}\}_{n=1}^{N}$, DiffEdit yields

$$E(t_{\text{pos}}) = \frac{1}{M}\sum_{m=1}^{M} E(s_{\text{pos},m} + o) + \text{DiffEdit}(t_{\text{pos}}),$$

$$E(t_{\text{neg}}) = \frac{1}{N}\sum_{n=1}^{N} E(s_{\text{neg},n} + o) + \text{DiffEdit}(t_{\text{neg}}),$$

two semantically disentangled prototypes anchored by both linguistic Δ_T and visual Δ_I guidance.

Directional Anomaly Score. Given query image x with embedding $E(x)$,

$$\text{score}(x) = \cos\big(E(x), E(t_{\text{neg}})\big) - \cos\big(E(x), E(t_{\text{pos}})\big),$$

where larger values imply stronger anomalous affinity. This soft differential metric preserves

fine-grained angular cues that hard thresholds miss. The score is later fused with segmentation evidence for added robustness.

3.7 AS in DE-CLIP

Following few-shot AD practice (Jeong et al., 2023; Li et al., 2024b), the AS module aggregates multi-scale CLIP features. For scales $l \in \{\text{low}, \text{mid}, \text{high}\}$, extract patches $\{\mathbf{v}_i^{(l)}(x)\}_{i=1}^{N_l}$ and build a normal memory bank $\mathcal{R}^{(l)} = \{\mathbf{r}_j^{(l)}\}_{j=1}^{K_l}$. The per-patch anomaly score is the minimum cosine distance

$$A_i^{(l)} = \min_j \big(1 - \cos(\mathbf{v}_i^{(l)}(x), \mathbf{r}_j^{(l)})\big),$$

and the cross-scale fusion uses a harmonic mean

$$A_i = \Big(\sum_l \tfrac{1}{A_i^{(l)}+\epsilon}\Big)^{-1},$$

which is then upsampled to an image-level anomaly map. This multi-resolution scheme captures both local defects and global structural deviations, entirely without pixel-level labels.

4 Experiments

4.1 Experiment Setups

Datasets. Following prior AD literature (Huang et al., 2024; Cao et al., 2024; Chen et al., 2023; Jeong et al., 2023; Zhou et al., 2023), we benchmark on two industrial datasets, MVTec AD (Bergmann et al., 2019) and VisA (Zou et al., 2022), and one medical dataset, BMAD Brain MRI (Bao et al., 2024a). Together they span natural photographs and MR images and provide both image- and pixel-level ground truth, allowing unified evaluation of detection and localisation.

Model and Training Protocol. DiffEdit is conceived as a training-free, plug-and-play module that drops into existing AD frameworks without any fine-tuning. For our main instantiation, DE-CLIP follows (Cao et al., 2024; Chen et al., 2023; Qu et al., 2024; Zhou et al., 2023) and uses OpenCLIP with a ViT-L/14 backbone as the visual encoder.

Metrics and Hyperparameters. Consistent with (Deng and Li, 2022; You et al., 2022; Defard et al., 2021; Roth et al., 2022; Huang et al., 2024; Cao et al., 2024; Chen et al., 2023; Jeong et al., 2023), we report Area Under the ROC Curve (AUROC), average precision (AP) and the maximum F1 score (max-F1) under the optimal threshold to evaluate both image-level and pixel-level AD performance. Unless stated otherwise, guidance weights are fixed to $\alpha = \beta = 0.5$ for every dataset, Sec. 4.4 analyses the full sweep.

Implementation Details. Consistent with (Jeong et al., 2023; Chen et al., 2023; Li et al., 2024a,b; Zhang et al., 2024), we supply a small pool of normal reference images at test time to instantiate DiffEdit. Every experiment is run under three random seeds (10, 42, 59) and averaged. All computations are performed on a single NVIDIA DGX A100 GPU.

4.2 Comparison with Other Methods

We benchmark DE-CLIP against a diverse suite of CLIP-based anomaly detectors grouped by training depth.

No-Training Baselines Include. (i) CLIP (Radford et al., 2021), which compares "A photo of a normal/anomalous [object]" prompts to query images; (ii) CLIP-AC (Radford et al., 2021), which averages normal and anomalous ImageNet-style templates; and (iii) WinCLIP(+) (Jeong et al., 2023), which uses reference images at inference but keeps both CLIP encoders frozen.

Shallow-training is represented by VAND (Chen et al., 2023), which adds and trains a lightweight adapter at the head, leaving the backbone fixed.

In-depth training includes AA-CLIP (Ma et al., 2025), which fits multiple projectors to align orthogonalised modalities; AdaCLIP (Cao et al., 2024), which learns prompts end-to-end; and AnomalyCLIP (Zhou et al., 2023), which jointly fine-tunes both vision and text encoders.

Tables 1, 2, and 3 show that DE-CLIP delivers state-of-the-art AUROC and other metrics for both AC and AS on MVTec, VisA, and Brain MRI across different shot counts. Despite being entirely training-free, it equals or surpasses shallowly and deep baselines, e.g., outperforms WinCLIP and rivals VAND/AA-CLIP on MVTec, while ranking top-tier in AC and first–second in AS on Brain MRI.

4.3 Visualization

Figure 6 highlights consistent localisation from DE-CLIP: defects are sharply isolated on both MVTec-Hazelnut and Brain MRI. VAND misses anomalies entirely, WinCLIP fires in irrelevant areas, and AnomalyCLIP oscillates, over-detecting hazelnut regions but missing key brain lesions. DiffEdit's heat-maps, however, align closely with

Method	Training Required?	AC(AUROC/AP/max-F1)					AS(AUROC/AP/max-F1)				
		0-Shot	2-Shot	4-Shot	8-Shot	10-Shot	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot
CLIP	No	76.8/75.4/76.7	–	–	–	–	47.5/12.6/8.7	–	–	–	–
CLIP-AC		61.8/60.5/62.2	–	–	–	–	51.3/14.2/9.5	–	–	–	–
WinCLIP		90.4/92.7/95.6	93.7/94.5/96.9	95.3/94.9/97.6	95.6/95.4/97.6	95.8/95.4/97.9	82.3/24.8/18.2	93.8/43.8/39.5	94.2/49.5/42.2	94.5/45.9/42.4	94.5/46.3/42.6
DE-CLIP (Ours)		85.9/90.2/93.4	91.2/93.4/94.5	94.0/93.7/95.5	96.6/94.7/96.6	95.1/94.8/96.2	94.2/50.6/48.1	95.8/51.9/50.0	96.2/54.8/53.2	96.6/55.7/54.2	96.7/55.2/53.9
VAND	Shallow Training	84.2/88.6/89.8	90.9/92.4/95.7	91.8/92.7/96	92.2/92.8/96.2	92.1/92.7/96.1	87.2/25.4/19.1	95.2/54.1/51.7	95.8/55.5/53.2	96/57.4/54.9	96.1/57.7/55.1
AA-CLIP	In-depth Training	84.6/89.1/92.9 (Full-Shot)					91.4/46.1/44.8 (Full-Shot)				
AdaCLIP		84.8/90.3/92.0 (Full-Shot)					89.2/43.9/41.2 (Full-Shot)				
AnomalyCLIP		91.6/92.5/95.9 (Full-Shot)					91.0/45.9/44.2 (Full-Shot)				

Table 1: Average performance comparison of various methods and DE-CLIP on the **MVTec** dataset under different shot settings. The best, second-best, and third-best results on AUROC are highlighted as first, second, and third.

Method	Training Required?	AC(AUROC/AP/max-F1)					AS(AUROC/AP/max-F1)				
		0-Shot	2-Shot	4-Shot	8-Shot	10-Shot	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot
CLIP	No	66.4/71.4/76.6	–	–	–	–	46.6/3.4/2.3	–	–	–	–
CLIP-AC		65.0/70.5/76.7	–	–	–	–	47.8/3.4/2.4	–	–	–	–
WinCLIP		75.5/78.2/78.7	83.4/82.5/84.6	84.1/82.4/85.8	86.0/83.8/87.1	86.4/83.9/87.8	73.2/9.0/5.4	95.1/23.9/16.6	95.2/25.3/17.7	95.3/25.8/18.2	95.2/25.8/17.8
DE-CLIP (Ours)		79.1/78.4/80.2	82.5/82.0/84.1	83.1/81.8/85.6	86.5/83.4/87.8	87.1/84.6/88.4	90.2/30.5/23.3	97.0/32.3/24.3	97.2/34.1/25.7	97.6/35.4/27.1	97.6/35.5/26.9
VAND	Shallow Training	78.0/76.9/80.3	80.9/79.6 /83.5	85.9/82.5/89.3	86.5/83.7/89.8	87.0/83.5/90.2	87.2/18.5/14.3	95.3/34.4/27.3	95.4 /34.4/27.9	95.7/34.9/28.3	95.7/35.2 /28.5
AA-CLIP	In-depth Training	82.7/81.5/83.4 (Full-Shot)					90.9/32.2/31.8 (Full-Shot)				
AdaCLIP		82.5/81.5/85.2 (Full-Shot)					95.6/36.4/32.0 (Full-Shot)				
AnomalyCLIP		82.1/81.8/83.5 (Full-Shot)					91.1/34.6/27.1 (Full-Shot)				

Table 2: Average performance comparison of various methods and DE-CLIP on the **VisA** dataset under different shot settings. The best, second-best, and third-best results on AUROC are highlighted as first, second, and third.

Method	Training Required?	AC(AUROC/AP/max-F1)					AS(AUROC/AP/max-F1)				
		0-Shot	2-Shot	4-Shot	8-Shot	10-Shot	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot
CLIP	No	38.8/42.2/40.5	–	–	–	–	68.3/3.2/1.7	–	–	–	–
CLIP-AC		37.8/42.2/38.2	–	–	–	–	67.5/3.2/1.6	–	–	–	–
WinCLIP		60.7/90.6/87.8	62.2/90.8/89.7	65.7/90.8/89.4	65.1/90.7/89.1	64.3/90.7/88.8	87.7/9.1/4.8	92.1/15.1/8.8	91.4/13.7/7.8	91.4/13.6/7.7	91.3/13.5/7.5
DE-CLIP (Ours)		54.3/88.6/86.5	75.2/91.2/91.9	73.6/91.0/91.0	72.8/91/90.5	71.7/90.9/90.2	87.7/9.2/5.0	92.6/16.4/10	91.9/14.6/8.6	92.0/14.6/8.4	91.9/14.4/8.3
VAND	Shallow Training	49.9/80.6/72.7	65.7/86.3/86.7	65.1/86.1/87.1	67.4/86.1/87.3	67.8/86.1/87.5	84.7/14.2/8.4	85.9/11.0/13.7	85.9/10.9/13.8	86.0/11.4/14.0	85.5/11.5/14.2
AA-CLIP	In-Depth Training	44.9/52.8/40.9 (Full-Shot)					89.9/13.5/11.4 (Full-Shot)				
AdaCLIP		64.4/91.0/90.3 (Full-Shot)					91.3/14.3/10.6 (Full-Shot)				
AnomalyCLIP		68.9/91.1/90.1 (Full-Shot)					88.1/10.9/11.1 (Full-Shot)				

Table 3: Average performance comparison of various methods and DE-CLIP on the **Brain MRI** dataset under different shot settings. The best, second-best, and third-best results on AUROC are highlighted as first, second, and third.

Method w/ DiffEdit	Training Required?	AC					AS				
		0-Shot	2-Shot	4-Shot	8-Shot	10-Shot	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot
CLIP	No	+9.2	+13.0	+13.6	+13.9	+14.1	+15.4	+22.5	+13.2	+9.0	+25.3
CLIP-AC		+24.6	+28.3	+28.8	+29.9	+30.1	+4.4	+9.3	+4.1	+22.4	+7.2
WinCLIP		+0.4	+0.1	+0.7	+2.1	+0.7	+0.0	+2.2	+2.0	+2.4	+2.2
VAND	Shallow Training	+0.2	+0.4	+0.2	+0.6	+0.2	+0.9	+0.0	+0.0	+0.9	+0.2
AA-CLIP	In-Depth Training	+0.4	+0.3	+0.4	+0.4	+0.4	+0.7	+0.6	+0.6	+0.6	+0.6
AdaCLIP		+0.2	+0.4	+0.2	+0.2	+0.4	+0.4	+0.4	+0.6	+0.2	+0.2
AnomalyCLIP		+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.1	+0.0	+0.1	+0.1

Table 4: Average improvement in AUROC (%) on the **MVTec** dataset achieved by combining DiffEdit with each baseline method.

Method w/ DiffEdit	AC					AS				
	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot	0-Shot	2-Shot	4-Shot	8-Shot	10-Shot
WinCLIP	+3.5	+7.9	+1.7	+2.1	+1.7	+2.3	+2.2	+2.0	+2.4	+2.2
VAND	+9.0	+15.1	+14.2	+14.4	+14.9	+4.3	+9.3	+9.6	+9.5	+10.0

Table 5: Average improvement in AUROC (%) on the **Brain MRI** dataset achieved by combining DiffEdit with each baseline method.

ground truth, evidencing robust suppression of spurious responses and faithful enhancement of true anomaly cues across domains.

4.4 Ablation study

Impact of Hyperparameter. We grid-searched $\alpha, \beta \in \{0.1{:}0.1{:}1.0\}$ (100 pairs). As Fig. 7 shows, AD AUROC on 8-shot stays above 0.95 across the plane, peaking at $\alpha \in [0.1, 0.3]$, $\beta = 0.2$ (0.969). Although the default $(0.5, 0.5)$ is not the global optimum on MVTec, it remains near-optimal and transfers unchanged to all datasets, confirming that a single fixed weighting delivers strong, domain-agnostic performance.

Impact of Prompt-Scale. Varying the number of antonym pairs K from 5 to 30 reveals a saturation curve. AUROC jumps from $91.2\% \pm 0.8$ at $K = 5$ to $93.2\% \pm 0.4$ at $K = 10$, rises only marginally

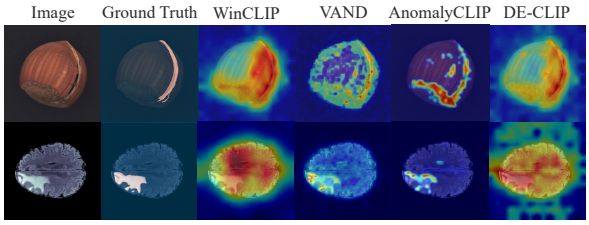


Figure 6: Visualization of anomaly localization results of different methods. DE-CLIP, WinCLIP, and VAND are evaluated under the 8-shot setting, while AnomalyCLIP uses the full-shot setting.

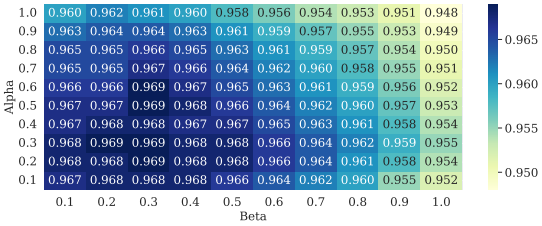


Figure 7: AUROC heatmap on MVTec across α and β.

to $93.4\% \pm 0.4$ at $K = 15$, and plateaus or dips slightly beyond $K \approx 20$. Thus, lexical diversity is beneficial up to a "sweet spot" of roughly 18 pairs, additional pairs introduce semantic noise that offsets the gains. Uniform and TF-IDF token weighting trace almost identical trajectories, indicating that once K is in the effective band, performance is governed more by the breadth of contrasts than by weighting heuristics.

Impact of Semantic Editing Components. The ablation results in Tab. 6 reveal that textual semantic editing and visual anchoring contribute orthogonally. The former enhances semantic separability, while the latter improves alignment with visual prototypes. Their combination introduces structured tension in the embedding space, enabling more robust anomaly detection without training.

Method	Avg. AUROC	
	AC	AS
Ours	96.6	95.4
w/o Δ_I	95.2 (-1.4)	89.2 (-6.2)
w/o $\Delta_{T(S)}$	92.3 (-4.3)	91.9 (-3.5)
w/o Δ	91.2 (-5.4)	87.8 (-7.6)

Table 6: Average AUROC (%) on MVTec in the 8-shot setting for the DE-CLIP and its three ablated variants.

$\Delta_{T(S+O)}$ vs $\Delta_{T(S)}$. We evaluate whether incorporating object-specific information in the difference vector is necessary by comparing DE-CLIP with $\Delta_{T(S+O)}$ and its simpler variant $\Delta_{T(S)}$, which uses only adjective-level semantics. As shown in Table 7, both variants achieve nearly identical performance on MVTec, with differences below 0.2% AUROC in both AC and AS tasks.

Method	Avg. AUROC	
	AC	AS
DE-CLIP (w/ $\Delta_{T(S+O)}$)	96.8	95.3
DE-CLIP (w/ $\Delta_{T(S)}$)	96.6	95.4

Table 7: Average AUROC (%) on MVTec in the 8-shot setting for $\Delta_{T(S+O)}$ and $\Delta_{T(S)}$ of DE-CLIP.

These results suggest that $\Delta_{T(S)}$ captures sufficient semantic contrast for guiding prompt editing, while offering a lightweight and generalizable alternative to object-conditioned editing.

4.5 Applying DiffEdit to Other Baselines

Table 4 assesses DiffEdit on MVTec by simply prepending its dual guidance $(\Delta_{T(S)}, \Delta_I)$ to each method's original prompt. A clear monotone emerges: the less a baseline has been fine-tuned, the more it gains. Training-free models jump by $> 30\%$ (AC) and $> 20\%$(AS); VAND, whose only learned component is a lightweight head—still improves, but deeply adapted variants (AA-CLIP, AdaCLIP, AnomalyCLIP) move by mere basis points. Extensive task-specific optimisation evidently solidifies CLIP's latent space, leaving little plasticity for external edits.

On Brain MRI (Table 5), DiffEdit likewise lifts VAND by $10 - 15\%$ across all shot counts and yields consistent boosts for WinCLIP, confirming that the guidance transfers from industrial to medical imagery.

Take-away. Increasing training depth is not invariably advantageous; fine-tuning on a few thousand images may disturb the parameter landscape learned from original CLIP and ultimately offer less headroom for simple yet effective prompt interventions like DiffEdit.

5 Conclusion

We propose DiffEdit, a dual-modality, plug-and-play embedding edit that sharpens CLIP's semantic boundaries, and built DE-CLIP, a training-free few-shot AD/AS framework. DiffEdit can be dropped into existing CLIP baselines to boost accuracy without retraining. Experiments on industrial and medical datasets show consistent gains in both classification and segmentation, demonstrating effectiveness and wide applicability in low-resource anomaly detection.

Limitations

Although DE-CLIP works without training, it presumes (i) a hand-crafted antonym lexicon for the textual vector Δ_T, which may be hard to define in domains such as satellite imagery or finance, and (ii) that the textual Δ_T and visual Δ_I point in similar directions, mis-alignment could weaken the edit, calling for adaptive or confidence-aware fusion. In addition, our anomaly-segmentation head reuses a classic sliding-window pipeline (Li et al., 2024b; Chen et al., 2023; Jeong et al., 2023; Ma et al., 2025) that overlooks long-range structure, hinting at future attention- or generation-based decoders.

Furthermore, many medical anomaly detection datasets provide only abnormal samples (Bao et al., 2024b; Bernal et al., 2012, 2015), making them incompatible with the proposed DiffEdit mechanism. Nevertheless, we believe that in most real-world scenarios, normal (positive) samples are substantially easier to collect than defective ones, and thus DE-CLIP remains practical and broadly applicable in realistic deployment settings.

Ethical Considerations

The primary objective of this work is to advance the understanding and safety of anomaly detection (AD) systems based on vision–language models (VLMs). All experiments are conducted using publicly available datasets, including MVTec AD, VisA, and BMAD Brain MRI, which are standard benchmarks widely adopted in the anomaly detection community. These datasets do not contain personally identifiable information or sensitive medical data beyond anonymized, research-grade imagery. Our method, DE-CLIP, operates in a training-free manner and does not require the collection of new data, thereby minimizing potential privacy or environmental concerns associated with large-scale model training.

This research does not pose foreseeable ethical risks such as privacy violations, bias amplification, or misuse. The proposed technique focuses on improving interpretability and robustness in few-shot anomaly detection rather than generating or modifying sensitive visual content. We will publicly release our implementation and experimental configurations to ensure transparency, reproducibility, and the responsible advancement of research in secure and trustworthy AD systems.

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A Linearizing the Hyperspherical Problem in VLM Embeddings

A.1 Ideal two-sided visual difference

Define the full visual shift

$$\Delta_I^{\pm} \;=\; \mu_+ - \mu_-. \tag{2}$$

Empirical collinearity. Because CLIP maximises cross-modal alignment, $\cos(\Delta_I^{\pm}, \Delta_T) \approx 1$. Hence there exists a constant $\delta > 0$ such that

$$\Delta_I^{\pm} \approx \delta\, \Delta_T. \tag{3}$$

Fisher Linear Discriminant. With the isotropy assumption $\Sigma_+ \approx \Sigma_- \approx \sigma^2 I$(Fisher, 1936), the Rayleigh quotient is maximised by

$$w^\star \;\propto\; \mu_+ - \mu_- \;=\; \Delta_I^{\pm}. \tag{4}$$

Impact of DiffEdit. DiffEdit injects a scaled visual shift:

$$\tilde{t}_{\text{pos}} = E(t_{\text{pos}}) + \beta \Delta_I^{\pm}, \qquad \beta \in [0, 1].$$

Because both anchors move by the same vector $\frac{\beta}{2}\Delta_I^{\pm}$,

$$\tilde{t}_{\text{pos}} - E(t_{\text{neg}}) = \Delta_T + \beta \Delta_I^{\pm} \overset{(3)}{\approx} (1+\beta\delta)\, \Delta_T. \tag{5}$$

Thus any β rescales but never rotates the optimal axis $w^\star$.

A.2 Practical one-sided visual anchor (normals only)

In real applications anomalous images are scarce, so we use the *one-sided* visual anchor

$$\Delta_I^{+} \;=\; \mu_+. \tag{6}$$

Collinearity still holds. CLIP alignment yields $\cos(\mu_+, E(t_{\text{pos}})) \approx 1$, and $E(t_{\text{pos}})$ is one endpoint of Δ_T; therefore

$$\cos(\Delta_I^{+}, \Delta_T) \;\geq\; \cos(\theta_{\max}) \approx 0.9,$$

i.e. the angle between Δ_I^{+} and Δ_T is at most $25°$. Hence there exists a constant $\gamma > 0$ such that

$$\Delta_I^{+} \approx \gamma\, \Delta_T. \tag{7}$$

DiffEdit with one-sided guidance. DiffEdit now becomes

$$\hat{t}_{\text{pos}} = E(t_{\text{pos}}) + \beta \Delta_I^{+}.$$

Its difference with respect to the negative anchor is

$$\hat{t}_{\text{pos}} - E(t_{\text{neg}}) = \Delta_T + \beta \Delta_I^{+} \overset{(7)}{\approx} (1+\beta\gamma)\, \Delta_T. \tag{8}$$

Thus, exactly as in the ideal case, injecting $\beta\Delta_I^{+}$ *only* scales the discriminative axis and never rotates it; experiments show that performance is most stable for $\beta \in [0.1,\, 0.3]$.

Both full-sided ($\Delta_I^{\pm}$) and one-sided (Δ_I^{+}) visual guidance satisfy the same collinearity property with Δ_T. Therefore, DiffEdit preserves the optimal FLD direction even when only a small set of normal reference images is available.